

1st REVIEW

Food Donation & NGO Connect App

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ABSTRACT

Large quantities of surplus food from restaurants and households go to waste every day, even as many people in the same neighbourhoods face food insecurity. A major barrier is the lack of a simple, trusted channel connecting willing donors with verified NGOs that can collect and redistribute this food quickly and safely. The Food Donation & NGO Connect App addresses this gap by providing a mobile platform, built on Android using Kotlin, that links restaurants and households with nearby NGOs for surplus food pickup. Donors can list available food items along with quantity, type, and pickup window through a simple Donor Listing feature. NGOs undergo an NGO Registration & Verification process to ensure only credible organizations participate, building trust within the platform. Once a donation is listed, the app uses Geo Matching, powered by the Google Maps SDK, to identify and notify the nearest verified NGOs, while a Pickup Scheduling module allows both parties to coordinate convenient collection times. A Notification Engine, implemented using Firebase Cloud Messaging (FCM), keeps donors and NGOs updated in real time on listing status, matches, and scheduling changes. All donation records are stored and managed using Firebase, ensuring reliable, real-time synchronization across users. Finally, an Impact Dashboard visualizes key metrics such as total food saved, number of successful pickups, and NGOs served, motivating continued participation. By streamlining the donation process end-to-end, the app aims to reduce food wastage, support NGOs with consistent food supply, and strengthen community-level food security.

INTRODUCTION

- Large amounts of edible surplus food from restaurants, catering events, and households are discarded daily, even as many people in the same neighbourhoods face food insecurity.
- NGOs working on food redistribution usually depend on phone calls, social media groups, or word of mouth to learn about available donations, which is slow and unreliable.
- This gap between food that would otherwise be wasted and organisations willing to redistribute it calls for a simple, trusted digital channel between donors and NGOs.
- The Food Donation & NGO Connect App is a mobile platform built on Android using Kotlin that links restaurants and households with nearby verified NGOs for surplus food pickup.
- The app combines donor listings, NGO verification, geo-matching, scheduling, and real-time notifications to make the donation process fast, transparent, and trustworthy.

EXISTING SYSTEM

- Food donation today largely relies on manual coordination — donors calling or messaging known NGO contacts individually with no structured process.
- General-purpose food delivery and e-commerce apps are not designed for donation logistics, verification, or time-sensitive pickup coordination.
- Some community WhatsApp/social media groups share donation information informally, with no verification of NGO credibility or guarantee of a timely response.
- There is no centralised system to match a donation with the nearest available NGO, so pickups are often delayed and food spoils before collection.
- Existing approaches offer no visibility into how much food is actually saved, no accountability, and do not scale beyond a small, personally known network.
- EXISTING APPLICATIONS : ShareTheMeal ,Careit , Food Rescue US , Karnaa

PROBLEM

- Donors lack a quick, reliable way to find a nearby NGO that can collect surplus food within its safe pickup window, so usable food is often thrown away.
- NGOs have no real-time visibility into when and where new donations become available, causing them to miss opportunities to collect food.
- There is no verification mechanism to confirm that an NGO receiving a donation is genuine and credible, which discourages some donors from participating.
- Coordinating pickup time and location manually is inefficient and error-prone, leading to missed collections and wasted effort on both sides.
- Without visibility into the impact of their contributions, donors and NGOs have little motivation to stay engaged with the process over time.

SOLUTION(Contribution)

An Android mobile application, built using Kotlin, that streamlines the entire donation lifecycle — listing, verification, matching, scheduling, notification, and tracking — in one platform.

- **Donor Listing:** a simple interface for donors to list available food items along with quantity, type, and pickup window.
- **NGO Registration & Verification:** ensures only credible, verified organisations can participate, building trust in the platform.
- **Geo Matching:** uses the Google Maps SDK to automatically identify and notify the nearest verified NGOs once a donation is listed.
- **Pickup Scheduling:** allows donors and NGOs to coordinate a convenient collection time and place.
- **Notification Engine:** built with Firebase Cloud Messaging (FCM) to keep both parties updated in real time on status, matches, and schedule changes.
- **Impact Dashboard:** visualises key metrics such as total food saved, successful pickups, and NGOs served to motivate continued participation.

Objectives

- To design and develop an Android application that connects surplus food donors with verified NGOs in real time.
- To implement a simple Donor Listing feature for capturing food type, quantity, and pickup window.
- To build an NGO Registration & Verification workflow that ensures only credible organisations participate.
- To implement Geo Matching using the Google Maps SDK for identifying and notifying the nearest available NGOs.
- To enable Pickup Scheduling so donors and NGOs can coordinate convenient collection times.
- To set up a real-time Notification Engine using Firebase Cloud Messaging (FCM) for status and schedule updates.
- To provide an Impact Dashboard that visualises food saved, pickups completed, and NGOs served.
- To reduce urban food wastage, support NGOs with a consistent food supply, and strengthen community-level food security.

Modules

- **1. Donor Listing Module:** lets restaurants and households list surplus food with quantity, type, and pickup window.
- **2. NGO Registration & Verification Module:** onboards NGOs and validates their credentials before granting platform access.
- **3. Geo Matching Module:** uses the Google Maps SDK to locate and notify the nearest verified NGOs for each new listing.
- **4. Pickup Scheduling Module:** coordinates collection date, time, and location between donors and NGOs.
- **5. Notification Engine:** sends real-time alerts via Firebase Cloud Messaging (FCM) for listings, matches, and schedule changes.
- **6. Data Management Module:** stores and synchronises all donation records in real time using Firebase.
- **7. Impact Dashboard Module:** visualises total food saved, successful pickups, and NGOs served.

SOFTWARE REQUIREMENTS

• Operating System	• Windows / macOS / Linux for development; Android 8.0 (Oreo) and above for the target device.
• IDE	• Android Studio.
• Programming Language	• Kotlin
• Backend & Database	• Firebase (Firestore / Realtime Database) for data storage and synchronisation.
• Cloud Messaging:	• Firebase Cloud Messaging (FCM) for real-time notifications.
• Maps & Location Services	• Google Maps SDK for Android for geo-matching.
• Version Control	• Git and GitHub.
• Design Tools	• Figma for UI/UX prototyping.

CONCLUSION AND FUTURE SCOPE

Conclusion

- The Food Donation & NGO Connect App gives restaurants, households, and NGOs a simple, trusted digital channel to redirect surplus food instead of discarding it.
- By automating NGO verification, geo-matching, scheduling, and notifications, the app reduces the delays that usually cause donated food to spoil before pickup.
- The Impact Dashboard makes the community-level impact of every donation visible, encouraging donors and NGOs to stay engaged over time.

Future Scope

- Extend the platform to iOS to widen the donor and NGO base beyond Android users.
- Introduce AI-based demand prediction to route donations toward the NGOs and areas with the greatest current need.
- Add volunteer or delivery-partner support for donors and NGOs unable to arrange collection themselves.
- Support multiple regional languages to make the app accessible to a wider range of donors and NGOs.
- Introduce recognition or reward features for consistent donors to further encourage long-term participation.

THANK YOU